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(54) **COMPRESSION WRAP WITH AN ARCHED SHAPE**

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(57) **ABSTRACT**

The invention relates to a compression bandage adapted to be wrapped around a body part in a helicoidal manner. The compression bandage has a length (L) and a width (W) and the length (L) is several times greater than the width (W). On a section of the compression bandage, the central longitudinal line (**15**) of the compression bandage has an arched shape.

COMPRESSION WRAP WITH AN ARCHED SHAPE

TECHNICAL FIELD

[0001] The invention relates to a compression bandage adapted to be wrapped around a body part in a helicoidal manner. The compression bandage has a length and a width and the length is several times greater than the width.

BACKGROUND ART

[0002] Multi-layer bandaging is the gold standard of care for lymphedema and other forms of edema. Multi-layer bandaging is typically applied in a clinical setting, with a trained therapist first wrapping the body part with a layer of cushioning (typically foam or non-woven cushioning). The cushioning acts to protect fragile skin, bridge small crevices or irregularities on the surface of the body part, and acts to provide even pressure on the skin when the multi-layer bandaging is applied. The therapist then wraps the body part with a short stretch bandage to provide the desired level of compression. The bandage is typically overlapped by approximately 50% so that two layers of bandaging are uniformly applied.

[0003] While the multi-layer bandaging process described above is highly effective, it consumes a lot of the therapist's time, limiting the number of patients they can see in a day. In many areas, the availability of trained therapists is limited, so many patients are not able to secure the treatment they need.

[0004] Another disadvantage of this solution is that because the treatment is so effective, the bandages may loosen quickly after application as the body part reduces in size. As a result, the dosage of compression applied is greatly reduced, or the bandages may completely fall off and provide no therapy at all. Since the application of the bandaging is complex and best performed by a trained therapist, the patient may simply remove the bandages until their next appointment, or misapply them and receive inadequate or improper treatment. In some countries, such as the US, patients typically receive treatment only twice a week, so proper treatment is lacking for significant periods of time.

[0005] Another disadvantage of multi-layer bandaging is that costs can be high, especially considering that the cushioning materials may be single use or have limited life and must be replaced frequently.

DISCLOSURE OF THE INVENTION

[0006] The problem to be solved by the present invention is to provide an alternative to the traditional multi-layer bandaging which can be more easily applied by the clinicians, therapists, home care providers or by the patient himself and that fits better the shape of a body part.

[0007] This problem is solved by a compression wrap in the shape of a bandage according to the independent claim. According to this, the compression bandage is adapted to be wrapped around a body part in a helicoidal manner. The compression bandage has a length and a width and the length is several times greater than the width. On a section of the compression bandage, the central longitudinal line of the compression bandage has an arched shape. I.e., when the bandage is laid out flat, its central lengthwise line is not straight but instead forms an arch.

[0008] The "central longitudinal line" corresponds to the line following the middle between the two outer edges of the section of the bandage. The "section" can be any section of the compression bandage.

[0009] The bandage initially was made without an arch, and the inventors discovered a tendency for a subsequent layer back face to not lay flat against the preceding layer's outer face. This was an indication that the subsequent layer was not providing the intended uniform compression on the preceding layer.

[0010] The solution was to incorporate a curve in the construction of the bandage so that it better fits the shape of a typical limb.

[0011] Advantageously, the section extends over at least one third of, preferably over at least half, preferably over the whole, length of the compression bandage. In particular, the section starts at one extremity of the compression bandage.

[0012] In a preferred embodiment, the arched shape is defined by a constant radius (R1) or a varying radius (R1, R2), wherein said constant radius (R1) or varying radius (R1, R2) is less than 4.0 meters, in particular less than 3.5 meters, in particular less than 3.0 meters, on the whole section.

[0013] In case the radius of the arch is constant, the arched shape has a radius R1 of

$$R_1 = r_1\sqrt{\frac{l^2}{(r_2 - r_1)^2} + 1}$$

wherein r1 is the radius of the start of the limb, r2 is the radius of the end of the limb, and I is the length of the limb.

[0014] In case of a varying radius (R1, R2), the varying radius may evolve linearly from a first radius (R1) to a second radius (R2), wherein the first radius (R1) corresponds to the radius at an extremity of the section and the second radius (R2) corresponds to the radius at the other extremity of the section. The resulting shape of the central longitudinal line is an Archimedean spiral.

[0015] Advantageously, the varying radius (R1, R2) evolves from a first radius R1 to a second radius R2, based on a constant equivalent to w/2π, in such a way that the shape of the arch is defined by

$$S_{plate}(\theta) = \begin{cases} x(\theta) = \frac{w}{4\pi}\theta\cos\left(\frac{r_2 - r_1}{\sqrt{l^2 + (r_2 - r_1)^2}}\theta\right) \\ y(\theta) = \frac{w}{4\pi}\theta\sin\left(\frac{r_2 - r_1}{\sqrt{l^2 + (r_2 - r_1)^2}}\theta\right) \\ z(\theta) = 0 \end{cases}$$

$$\begin{cases} \theta_0 = \frac{4\pi}{w}\cdot r_1 \cdot \frac{\sqrt{l^2 + (r_2 - r_1)^2}}{r_2 - r_1} \\ \theta_f = \frac{4\pi}{w}\cdot r_2 \cdot \frac{\sqrt{l^2 + (r_2 - r_1)^2}}{r_2 - r_1} \end{cases}$$

[0016] In particular, the second radius (R2) is larger, in particular 10% larger, in particular 20% larger, in particular 50% larger, than the first radius (R1).

[0017] Limbs tend to narrow at their extremities, such as ankles or wrists, and gradually widen towards the upper sections, including the calf, thigh, or the area approaching the axilla. This anatomical pattern is observable in various types of limbs. The arch of the compression bandage may be designed to accommodate the increase in circumference along the length of the limb.

[0018] The arched shape of the compression bandage may be formed by cutting the materials in an arch.

[0019] In the preferred embodiment, the compression bandage comprises and in particular consists of a first strip and a second strip, which are attached side-by-side, preferably by sewing. The first strip corresponds to a first part and the second strip corresponds to a second part. The first part extends across a part of the width of the compression bandage and the second part extends across a part of the remaining width of the compression bandage.

[0020] Alternatively to cutting the materials in an arched shape, in the preferred embodiment, the arched shape is created by stretching the second part relative to the first part, wherein the second part is situated closer to the concave side of the compression bandage than the first part.

[0021] In another embodiment, if the bandage is comprised of a knitted or woven material, the tension in the yarns may be adjusted so that the bandage does not lay straight, but rather curves.

[0022] In a preferred embodiment, the length of the compression bandage is at least six times, preferably at least eight times, preferably at least ten times, greater than the width of the compression bandage. In particular, the length of the compression bandage is greater than 60 cm, preferably greater than 80 cm, preferably greater than 100 cm.

[0023] Advantageously, the compression bandage comprises a starting portion and a central portion, wherein the central portion is adjacent to the starting portion. The starting portion and the central portion comprise the first part and the second part, both extending at least over the entire length of the central portion. The first part has a first thickness and a first width and the second part has a second thickness and a second width, wherein at any point along its length, in the starting portion and in the central portion, the second width is at least half the first width and the first thickness is bigger than the second thickness.

[0024] When wrapping the compression bandage around the body part, the first part acts as a first cushioning layer over the body part. The bandage is wrapped in such a way that the second part covers at least partly the first part of the previous wrap to provide a further compressive effect over the cushioning layer.

[0025] In particular, the central portion is at least 70%, preferably at least 80%, preferably at least 90%, of the length of the compression bandage and/or the length of the central portion is at least 50 cm, preferably at least 70 cm, preferably at least 90 cm.

[0026] In a preferred embodiment, in the starting portion and in the central portion, the first part comprises a cushioning material. Given that the compression bandage offers both cushioning and compression within a single layer, the time required for application is significantly reduced.

[0027] In particular, in the starting portion and in the central portion, the first part consists only of the cushioning material.

[0028] Alternatively, in the starting portion and in the central portion, the first part comprises at least a layer of the cushioning material. This layer extends over the whole length of the first part. In particular, the layer of the cushioning material extends over at least the half of the thickness of the first part.

[0029] In particular, in the starting portion and in the central portion, the cushioning material extends over the whole width of the first part.

[0030] In a preferred embodiment, the cushioning material is open or closed cell foam, laminated open or closed cell foam, non-woven cushioning or woven cushioning preferably open cell foam or laminated open cell foam, preferably laminated open cell foam.

[0031] Open cell foam and laminated open cell foam are advantageous, as they enable a good air flow, moisture and vapor transmission from the patient body through the compression bandage.

[0032] The laminate layer present on the upper surface and in particular present on the upper surface of the first part, can be any material known in the art suitable for bandaging a body part including natural and synthetic fibers. Examples could include 100% cotton highly twisted yarns and synthetic fibers including elastane yarns to provide stretch. In a preferred embodiment, the laminate layer present on the upper surface of the first part is a hook receptive nylon/elastane loop fabric suitable for flame lamination. The lower surface of the compression bandage corresponds to the surface facing the skin when wrapped around the body part and the upper surface corresponds to the opposite surface, not facing the body part.

[0033] The laminate layer present on the lower surface and in particular, present on the lower surface of the first part, may be any material known in the art suitable for contact with the skin, including natural and synthetic fibers.

[0034] Advantageously, the starting portion comprises an extension of the first part protruding past the end of the second part.

[0035] In the first wrap of bandaging that encircles the body part once, serving as the starting point for subsequent wraps, the compression bandage has no previous wrap to lay against. Consequently, if both the first part and the second part were to initiate from the same position along its length, an extra piece of useless second part would be hanging. This hanging extra piece could hinder the proper application of the compression bandage at the necessary low position. For instance, if the compression bandage would be applied to the calf, starting from the ankle area, it would be needed to start applying the compression bandage slightly above the ankle for the hanging extra piece to not bother the foot.

[0036] Thus, it is of interest to start the second part after the first wrap of bandaging of the first part around the body part. This results in a design where the starting portion has an extension of the first part protruding past the end of the second part.

[0037] The compression bandage, as sold, should have an extension longer than the circumference of the body part such that the extension can be trimmed to the right size to encircle the body part in a customized manner.

[0038] Advantageously, the extension is at least 4%, preferably at least 6%, preferably at least 10% of the length of the compression bandage. In particular, the extension of the first part is longer than 9 cm, preferably longer than 12 cm, preferably longer than 15 cm. Such a length corresponds at least to one circumference of the limb to be wrapped.

[0039] In particular, at any point along its length, in the starting portion and in the central portion, the second width is at least 60% of the first width, preferably at least equal to the first width. The first width can be between 2 to 8 cm, in particular between 3 to 6 cm, and the second width can be between 2 to 9 cm, in particular between 3 to 7 cm.

[0040] Advantageously, at any point along its length, in the starting portion and in the central portion, the first thickness is at least 25% greater, preferably at least 50% greater, preferably at least 100% greater than the second thickness.

[0041] In particular, in the starting portion and in the central portion, the first thickness is between 3 to 10 mm, in particular between 5 to 7 mm, and/or the second thickness is between 1 to 5 mm, in particular between 2 mm to 3 mm.

[0042] In a preferred embodiment, in the starting portion and in the central portion, the first width is essentially constant, and/or the second width is essentially constant. In particular, in the starting portion and in the central portion, the first thickness is essentially constant and/or the second thickness is essentially constant.

[0043] An "essentially" constant thickness means that the thickness of the first part and of the second part are mostly constant and it is not excluded that the thickness is locally different, e.g., due to locally arranged tabs of loop-material and due to manufacturing tolerances. In particular, at least 80%, in particular at least 90% of the first part and/or of the second part have an identical thickness.

[0044] In the same way, an "essentially" constant width means that the width of the first part and of the second part are mostly constant and it is not excluded that the width is locally different, e.g., due to manufacturing tolerances. In particular, at least 80%, in particular at least 90% of the first part and/or of the second part have an identical width.

[0045] Advantageously, in the starting portion and in the central portion, the first part extends across a part of the width of the compression bandage and the second part extends across the remaining part of the width of the compression bandage.

[0046] In particular, in the starting portion and in the central portion, the first part and the second part consist of a first single continuous piece and a second single continuous piece. The first single continuous piece and the second single continuous piece are distinct. The first single continuous piece and the second single continuous piece are distinct in the sense that they do not have joins, seams, or separate parts. They are made of one bloc without any additional pieces attached or assembled. Each single continuous piece may be made of one or multiple materials (e.g. in foam laminate).

[0047] In particular, the first part consists of the first single continuous piece and the second part consists of the second single continuous piece.

[0048] The first single continuous piece is preferentially open or closed cell foam, laminated open or closed cell foam, non-woven cushioning or woven cushioning, preferably open cell foam or laminated open cell foam, preferably laminated open cell foam.

[0049] The laminated open cell foam of the first single continuous piece may have a density of 1 to 10 lb./ft^3, which corresponds to a density of 16 to 160 kg/m^3.

[0050] Preferentially, the second single continuous piece may also be laminated open cell foam, in particular in a compressed form. The laminated open cell foam or the laminated compressed open cell foam of the second single continuous piece may have a density of 3 to 12 lb./ft^3, which corresponds to a density of 48 to 192 kg/m^3.

[0051] Other advantageous embodiments are listed in the dependent claims as well as in the description below.

[0052] The invention will be better understood and objects other than those set forth above will become apparent from the following detailed description thereof. Such description makes reference to the following:

[0053] Embodiments describe a compression bandage according to the invention; A first embodiment of the compression bandage; A second embodiment of the compression bandage; A third embodiment of the compression bandage; A wrapping of the compression bandage around the calf; A truncated cone for explaining the geometry of the compression bandage; An unrolled truncated cone; and Arches of two different compression bandages, dimensioned for different body parts.

MODES FOR CARRYING OUT THE INVENTION

[0054] Embodiments of a compression bandage according to the invention. The compression bandage is adapted to be wrapped around a body part, in particular a calf, a foot, a thigh or an arm. The compression bandage has a length and a width.

[0055] The compression bandage consists of a starting portion, a central portion and a terminal portion. For explanatory purposes, these three portions are separated by a central longitudinal line. The central portion is arranged between the starting portion and the terminal portion. I.e., the central portion is adjacent to the starting portion and the central portion is adjacent to the terminal portion.

[0056] The compression bandage consists of two parts. The starting portion, the central portion and the terminal portion comprise a first part and a second part. The first part comprises an extension protruding past the end of the second part. I.e., the first part extends over the whole length of the compression bandage while the second part is shorter. The extension is arranged in the starting portion. The central longitudinal line corresponds to the boundary between the first part and the second part.

[0057] In the exemplary embodiment, the length of the extension is 35 cm. The extension is longer than the circumference of the body part to be wrapped such that it can be trimmed to the right size.

[0058] The first part has the same width as the second part.

[0059] Both, the first part and the second part of the compression bandage, as sold, have square ends. It is recommended for the therapist to trim the ends of the first part and of the second part with scissors. Advantageously, the strips of the first part and of the second part consist of materials that do not fray after being cut.

[0060] In all three embodiments, the first part has a first thickness, a first width, and the second part has a second thickness and a second width. The first width of the first part and the second width of the second part are identical, i.e., both, the first part and the second part each extend over 50% of the width of the compression bandage, collectively encompassing the whole width. In an exemplary embodiment, the first thickness is 5 mm, the second thickness is 2.5 mm, both the first and second widths are each 4 cm, i.e., the total width is 8 cm. The length of the compression bandage can be 5 meters.

[0061] The compression bandage has a lower surface facing the skin when wrapped around the body part and an opposite upper surface, not facing the body part. The upper surface is flat. The lower surface is not flat, since the thickness of the compression bandage decreases from the first thickness to the second thickness in the middle of the total width of the compression bandage.

[0062] The first embodiment consists of only one single continuous piece. The first part and the second part are made from the same single continuous piece.

[0063] Alternatively, as provided in the second embodiment the compression bandage consists of two single continuous pieces that have been layered. The second single continuous piece extends over the width of the compression bandage and has a constant thickness corresponding to the second thickness. The first single continuous piece extends over half of the width of the compression bandage and is placed on the lower surface of the second single continuous piece in such a way that a side of the first single continuous piece aligns with a side of the second single continuous piece. The first single continuous piece has a thickness corresponding to the difference between the first thickness and the second thickness. As such, the first part consists of a layer of the first single continuous piece and of a layer of the second continuous piece whilst the second part consists only of the second continuous piece.

[0064] Regarding the third embodiment, the first single continuous piece extends over half of the width of the compression bandage and has a constant thickness corresponding to the first thickness. The second single continuous piece extends over the other half of the width of the compression bandage and has a constant thickness corresponding to the second thickness. As such, the first part consists only of the first single continuous piece and the second part consists only of the second single continuous piece. Preferentially, the first single continuous piece and the second single continuous piece are attached side by side by sewing.

[0065] The third embodiment is preferred as in this embodiment, the arched shape can be created by stretching the second single continuous piece relative to the first single continuous piece preferably during the sewing process. The arch is created when the second single continuous piece that was stretched, contracts and forms the concave side of the arch.

[0066] In this third embodiment, the first single continuous piece and the second single continuous piece are attached side by side by sewing, in particular by zig-zag stitches bar tacked in at least one location to prevent the stitches from unravelling so that the length of the compression bandage may be cut to length in at least one location. Alternatively, the two pieces can be ultrasonically welded or otherwise secured by tape, adhesives, or any other suitable means.

[0067] In all three embodiments, the first single continuous piece can be made of open cell foam or laminated open cell foam. The second single continuous piece can be made of compressed open cell foam or laminated compressed open cell foam.

[0068] Regarding how the compression bandage is wrapped around a calf. The ends of the compression bandage have already been trimmed with scissors to the desired length and shape. In a first step, the starting portion, is wrapped around the ankle. The length of the extension after trimming matches the circumference of the ankle.

[0069] After having wrapped the extension of the first part around the calf, the second part starts and the width of the compression bandage doubles. In the following wraps, the second part of the compression bandage covers the first part.

[0070] The wraps are continued until the calf is covered to the knee. The tab of hook-material is arranged on the end of the terminal portion for securing the end of the compression bandage. The upper layer of the first part and the upper layer of the second part are both loop-material. Half of the tab of hook-material can be attached on the upper surface of the terminal portion, whilst the other half is attached to the upper surface of the first part of the previous wrap.

[0071] Furthermore, other tabs of hook-material are arranged on the lower surface of the second part. When wrapping the compression bandage around the body part, the loop-material of the upper surface of the first part connects with the tabs of hook-material of the lower surface of the second part.

[0072] The compression bandage can be wrapped around the calf. The starting portion is arranged at the ankle. Five more wraps of the central portion follow and at the end of the compression bandage. Tabs of hook-material are arranged at the lower surface of the compression bandage for progressively securing the compression bandage as it is being wrapped around the calf.

[0073] The punctual increase in thickness from the second thickness to the first thickness creates a ridge that enables. The compression bandage disclosed provides both a ridge to guide and support each subsequent wrap of the compression bandage. In the third embodiment in which the first single continuous piece and the second single continuous piece are attached side by side by sewing, the sewn line that further provides visual guidance. In this way, it is easy to apply the helicoidal wraps with consistent overlap for both a trained therapist and the patient or home care provider.

[0074] The ridge allows each succeeding helicoidal wrap is mechanically supported by the preceding layer, supporting the compression bandage in place as the body part reduces in size in response to the therapeutic compression applied. Periodically located tabs of hook-material provided on the lower surface of the second part attach to the upper surface of the first part of the preceding layer. In this way, the hook and loop fasteners provide a second means to hold the compression bandage in place even if the body part reduces in circumference.

[0075] The tabs of hook-material provide periodic anchoring points to maintain the compression bandage in place whilst, increasing the ease of application for the therapist, the home care provider, or the patient himself.

[0076] Due to the ease of application and guidance provided by the ridge of the first part and sewn line to properly locate each compression bandage spiral layer, the compression bandage can easily and reliably be reapplied by the patient or home care provider as the body part circumference reduces in response to the therapeutic compression applied. In this way, proper therapeutic compression can be maintained between clinical sessions.

[0077] Thus, the present solution provides a one-piece solution that provides an alternative to the multi-layer bandaging that can be quickly and easily applied by trained professionals, home care providers, and patients.

[0078] As already mentioned, the compression bandage i comprises an arched shape, or more specifically, the central longitudinal line has an arched shape. The arch is defined by its constant or varying radius. The constant or the varying radius can be derived by unrolling a truncated cone, because a truncated cone approximates the shape of a body part, such as a calf.

[0079] The truncated cone is defined by a smaller radius r1, a bigger radius r2 and a length. If this geometry approximates a calf, the double of the radius r1 would correspond to the diameter of the ankle and the double of the radius r2 would correspond to the diameter of the calf below the knee, at its widest point.

[0080] When unrolled, the truncated cone is defined by a first radius R1 and a second radius R2.

[0081] The radius R1 corresponds to

$$R_1 = r_1\sqrt{\frac{l^2}{(r_2 - r_1)^2} + 1}$$

[0082] In case of a compression bandage with a constant radius, we use the inner radius R1 as the radius R of the central longitudinal line of the compression bandage.

[0083] The compression bandage would fit even better to a body part approximated by the truncated cone, if the central longitudinal line of the compression bandage had a radius R evolving from smaller radius R1 to a bigger radius R2. In case the body part is a calf, the one end of the compression bandage with the smaller radius R1 would be wrapped around the ankle and the other end of the bandage with the bigger radius R2 would be wrapped around the calf below the knee.

[0084] A compression bandage with a varying radius fits best a body part if the varying radius evolves linearly from a first radius R1 to a second radius R2 which is defined as follows:

$$S_{plate}(\theta) = \begin{cases} x(\theta) = \frac{w}{4\pi}\theta\cos\left(\frac{r_2 - r_1}{\sqrt{l^2 + (r_2 - r_1)^2}}\theta\right) \\ y(\theta) = \frac{w}{4\pi}\theta\sin\left(\frac{r_2 - r_1}{\sqrt{l^2 + (r_2 - r_1)^2}}\theta\right) \\ z(\theta) = 0 \end{cases}$$

$$\begin{cases} \theta_0 = \frac{4\pi}{w} \cdot r_1 \cdot \frac{\sqrt{l^2 + (r_2 - r_1)^2}}{r_2 - r_1} \\ \theta_f = \frac{4\pi}{w} \cdot r_2 \cdot \frac{\sqrt{l^2 + (r_2 - r_1)^2}}{r_2 - r_1} \end{cases}$$

[0085] The resulting shape is an Archimedean spiral.

[0086] Embodiments may have two different Archimedean spirals. The radius of arch of the Archimedean spirals evolves linearly from a smaller radius R1 to a bigger radius R2. The arches correspond to the central longitudinal line of the compression bandage.

[0087] The different bandages can be designed for different body parts and for people of different sizes. The dimensions could be as follows:

[0088] for a calf, the width of the bandage can vary from 6 to 12 cm; preferably the bandage has a width of 8 to 10 cm; preferably 8 cm; the length of the bandage can vary from 1.20 to 7.00 m depending on width; For a narrow bandage width of 8 cm, its length can be 1.80 to 5.30 m; for a wide bandage width of 10 cm, its length can be 1.40 to 4.30 m.

[0089] for a foot, the width of the bandage can vary from 4 to 10 cm; preferably the bandage has a width of 6 to 8 cm; preferably 6 cm, the length of the bandage can vary from 0.70 to 5.20 m depending on width; For a narrow bandage width of 6 cm, its length can be 1.20 to 3.50 m; for a wide bandage width of 8 cm, its length can be 0.90 to 2.70 m.

[0090] for a thigh, the width of the bandage can vary from 8 to 14 cm; preferably the bandage has a width of 10 to 12 cm; preferably 12 cm, the length of the bandage can vary from 1.20 to 7.50 m depending on width; For a narrow bandage width of 10 cm its length can be 1.70 to 6.30 m; For a wide bandage width of 12 cm, its length can be 1.50 to 5.30 m.

[0091] for an arm, the width of the bandage can vary from 4 to 10 cm; preferably, the bandage has a width of 6 to 8 cm; preferably 6 cm, the length of the bandage can vary from 1.50 to 10.00 m depending on width; For a narrow bandage width of 6 cm, its length can be 2.40 to 6.70 m; For a wide bandage width of 8 cm, its length can be 1.80 to 5.10 m.

[0092] the bandage may be trimmed at multiple locations to best match the circumference and the length of the limb being wrapped.

1. A compression bandage adapted to be wrapped around a body part in a helicoidal manner, wherein the compression bandage has a length (L) and a width (W) and the length (L) is several times greater than the width (W), wherein on a section of the compression bandage, the central longitudinal line (**15**) of the compression bandage has an arched shape.

2. The compression bandage according to claim **1**, wherein the section extends over at least one third of, preferably over at least half, preferably over the whole, length (L) of the compression bandage.

3. The compression bandage according to claim **1**, wherein the compression bandage comprises, in particular consists of a first part (4) and a second part (5) attached side by side, wherein the first part (4) and the second part (5) extend at least over a part of the length of the compression bandage, the first part (4) extending across a part of the width (W) of the compression bandage and the second part (5) extending across a part of the remaining width of the compression bandage, wherein the arched shape is created by stretching the second part (5) relative to the first part (4), wherein the second part (5) is situated closer to the concave side of the compression bandage than the first part (4).

4. The compression bandage according to claim **1**, wherein the section starts at one extremity of the compression bandage.

5. The compression bandage according to claim **1**, wherein the arched shape is defined by a constant radius (R) or a varying radius (R_1, R_2), wherein said constant radius (R) or varying radius (R_1, R_2) is less than 4.0 meters, in particular less than 3.5 meters, in particular less than 3.0 meters, on the whole section.

6. The compression bandage according to claim, **1**, wherein the arched shape is defined by a varying radius (R_1, R_2), wherein the varying radius (R_1, R_2) evolves linearly from a first radius (R_1) to a second radius (R_2), wherein the first radius (R_1) corresponds to the radius at an extremity of the section and the second radius (R_2) corresponds to the other extremity of the section.

7. The compression bandage according to claim **6**, wherein the first radius (R_1) corresponds to the extremity of the compression bandage, wherein the second radius (R_2) is larger, in particular 10% larger, in particular 20% larger, in particular 50% larger, than the first radius (R_1).

8. The compression bandage according to claim **1**, wherein the length (L) of the compression bandage is at least six times, preferably at least eight times, preferably at least ten times greater than the width (W) of the compression bandage.

9. The compression bandage according to claim **1**, wherein the length (L) of the compression bandage is greater than 60 cm, preferably greater than 80 cm, preferably greater than 100 cm.

10. The compression bandage according to claim **1**, wherein

the compression bandage comprises a starting portion (1) and a central portion (2), wherein the central portion (2) is adjacent to the starting portion (1),

the starting portion (1) and the central portion (2) comprise the first part (4) and the second part (5), both extending at least over the entire length of the central portion (2), wherein the first part (4) has a first thickness (T_1) and a first width (W_1) and the second part (5) has a second thickness (T_2) and a second width (W_2), wherein at any point along its length, in the starting portion (1) and in the central portion (2), the second width (W_2) is at least half the first width (W_1) and the first thickness (T_1) is bigger than the second thickness (T_2).

11. The compression bandage according to claim **9**, wherein the central portion (2) is at least 70%, preferably at least 80%, preferably at least 90%, of the length of the compression bandage.

12. The compression bandage according to claim **9**, wherein the length of the central portion (2) is at least 50 cm, preferably at least 70 cm, preferably at least 90 cm.

13. The compression bandage according to claim **9**, wherein in the starting portion (1) and in the central portion (2), the first part (4) comprises a cushioning material.

14. The compression bandage according to claim **11**, wherein in the starting portion (1) and in the central portion (2), the first part (4)

consists of the cushioning material, or

comprises at least a layer of the cushioning material extending over the whole length of the first part (4), in particular wherein

the layer extends over at least half of the thickness (T_1) of the first part (4).

15. The compression bandage according to claim **14**, wherein the cushioning material extends over the width of the first part (4).

16. The compression bandage according to claim **11**, wherein the cushioning material is open or closed cell foam, laminated open or closed cell foam, non-woven cushioning or woven cushioning, preferably open cell foam or laminated open cell foam, preferably laminated open cell foam.

17. The compression bandage according to claim **9**, wherein the starting portion (1) comprises an extension of the first part (4) protruding past the end of the second part (5).

18. The compression bandage according to claim **6**, wherein the first radius (R_1) corresponds to the extremity of the compression bandage of the starting portion (1).

19. The compression bandage according to claim **17**, wherein the extension of the first part (4) is at least 4%, preferably at least 6%, preferably at least 10% of the length of the compression bandage.

20. The compression bandage according to claim **17**, wherein the extension of the first part (4) is longer than 9 cm, preferably longer than 12 cm, preferably longer than 15 cm.

21.-**24**. (canceled)

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